

Project Planning Phase

Project Planning (Product Backlog, Sprint Planning, Stories, Story Points)

Field	Details
Team Lead	Govindu Kavya Sri
Team Members	Eswari Gayatri Budumuru, Kiladi Kalavathi, Adaka Mounika, Panditi Jyostna
Domain	Retail / Cosmetics Analytics
Tools Used	Python, Microsoft Excel, Tableau Public
Dataset	Cosmetics product dataset — 1,472 records, 11 attributes
Published Dashboard	public.tableau.com/app/profile/govindu.kavya.sri/viz/cosmeticproject_17872016666220/Dashboard1

Project Context

The cosmetics industry is one of the fastest-growing and highly competitive sectors, driven by changing consumer preferences, diverse skin types, brand innovation, and evolving pricing strategies. With increasing demand for personalized beauty products, companies rely on data-driven insights to understand market trends, consumer behavior, and product performance.

This project, “Cosmetic Insights: Navigating Cosmetics Trends and Consumer Behavior with Tableau,” focuses on analyzing a cosmetics dataset to uncover meaningful patterns and actionable insights. The dataset includes key attributes such as brand, product category, price, rank, and skin type suitability (Dry, Oily, Combination, Normal, and Sensitive).

This project is designed to help businesses and analysts understand cosmetics market trends and consumer preferences. By analyzing brand performance, pricing patterns, product distribution, and skin-type coverage, the dashboard enables stakeholders to identify high-performing products, target specific customer segments, and optimize marketing and product strategies.

Scenario 1: Monitoring Consumer Preferences

In a real-time scenario, imagine receiving an alert indicating a concerning trend in consumer preferences, such as a significant decline in interest in certain cosmetic products or ingredients. Using the Cosmetic Insights data, we can promptly assess the extent and potential impact of this trend, identify contributing factors, and deploy immediate interventions to adapt product offerings and marketing strategies. Whether

it's through targeted promotional campaigns, adjustments in product formulations, or personalized recommendations, real-time analysis enables agile decision-making and proactive measures to meet evolving consumer needs.

Scenario 2: Addressing Product Concerns

In the event of identifying widespread product concerns, such as negative reviews or safety issues associated with specific cosmetic items, real-time access to Cosmetic Insights data enables swift response and management. Cosmetic companies and regulatory bodies can utilize the dataset to gather crucial information about the concerns, including their prevalence, potential impacts on consumer trust, and affected product demographics. By leveraging real-time analytics, they can implement quality control measures, recall products if necessary, and communicate transparently with consumers to address their concerns and maintain brand integrity.

Scenario 3: Predictive Analysis and Product Innovation

Leveraging predictive analytics capabilities, Cosmetic Insights empowers companies to anticipate and respond to emerging trends and consumer preferences in the beauty industry. By analyzing historical data and identifying predictive indicators, companies can proactively innovate new products, adjust existing formulations, and tailor marketing strategies to meet evolving consumer demands. Real-time monitoring of market trends, consumer feedback, and competitor activities enables timely interventions, product innovation, and strategic decision-making to stay ahead in a competitive market landscape.

Product Backlog & Sprint Schedule

The tasks below were carried out by the project team across three sprints, coordinated by the team lead, and organized around the eight epics tracked on the SkillWallet platform (8 epics, 14 stories/tasks, 8 subtasks).

Epic	Task	Points	Priority	Sprint
Data Collection	Source and load the cosmetics dataset (1,472 rows)	2	High	Sprint 1
Data Collection	Inspect structure, nulls, and duplicates	1	High	Sprint 1
Data Preparation	Clean brand casing, ingredient placeholders, zero-ranks using Python Pandas	3	High	Sprint 1
Data Preparation	Reformat and export cleaned Excel file; connect to Tableau	2	High	Sprint 1
Data Visualization	Build Avg Price by Category and Price vs Rating charts	3	High	Sprint 2
Data Visualization	Build Top Brands by Category chart (Top 10, sorted)	2	High	Sprint 2
Data Visualization	Pivot skin-type columns; build Skin Type Coverage heatmap	3	Medium	Sprint 2
Dashboard	Assemble all four charts into a single dashboard	3	High	Sprint 2

Epic	Task	Points	Priority	Sprint
Dashboard	Add cross-chart filter actions and searchable Brand filter	3	High	Sprint 2
Story	Draft narrative structure connecting dashboard views to consumer, retailer, and analyst scenarios	2	Medium	Sprint 3
Performance Testing	Test all filters for responsiveness; document load time and data volume	2	Medium	Sprint 3
Web Integration	Publish workbook to Tableau Public; verify URL access	2	High	Sprint 3
Documentation	Complete all phase documentation deliverables and final report	5	High	Sprint 3
Documentation	Capture dashboard screenshots and prepare demo walkthrough	3	High	Sprint 3

Velocity

Sprint 1 = 8 points | Sprint 2 = 14 points | Sprint 3 = 14 points | Total = 36 story points. Sprint 2 carried the heaviest per-task load because it covered both chart-building and dashboard assembly — the most integration-heavy deliverables — while Sprint 3's task count reflects the breadth of publishing, testing, and documentation work needed to close out the project.